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MILITARY SPECIFICATION

COMPOSITION, DELAY

1. SCOPE

1.1 This specification covers zirconium-nickel alloy delay compositions intended for use in delay elements of fuzes.

1.2 **Classification.** The delay compositions shall be of the following types as specified (see 6.1):

Type I	- Delay Composition, 2-second
Type II	- Delay Composition, 5-second
Type III	- Delay Composition, 12-second

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366	- Sieves: Standard, Testing

MILITARY

MIL-P-217	- Potassium Perchlorate
MIL-B-550	- Barium Chromate (For Pyrotechnic Mixtures)
MIL-Z-11410	- Zirconium Nickel Alloy, Powdered

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

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MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1233	- Procedures for Determining Particle Size, Particle Size Distribution, and Packed Density of Powdered Material
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

7548570	- Fuze, Grenade, Hand, M204A2, M206A2 and Fuze, Grenade, Hand, Practice, M205A2 Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publications of the issue in effect on date of invitation for bids, form a part of this specification:

Code of Federal Regulations

Title 49	- Transportation: Chapter I - Interstate Commerce Commission, PArts 71 - 90 - Explosives and Other Dangerous Articles

(Parts 71 - 90 of Title 49 of the Code of Federal Regulations are contained in a volume which comprises parts 71 - 90 of that title. Copies of this volume may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Officer, Washington, D.C. Applications should cite "49 CFR 71 - 90.")

3. REQUIREMENTS.

3.1 **Materials.**

3.1.1 The raw materials used in the manufacture of delay composition purchased under this specification shall conform to the requirements of the applicable specifications listed in 2.1. In addition, the following materials shall be of the type, grade or class specified as follows:

Barium chromate, MIL-B-550, Class A

Potassium perchlorate, MIL-P-217, Granulation IV, (see 4.5)

Zirconium-nickel alloy, powdered, MIL-Z-11410, Type I & II

3.2 **Composition.**

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3.2.1 The delay composition shall be manufactured in accordance with the following formulae:

Component	Type I	Type II	Type III
Barium Chromate, percent	60.0	60.0	60.0
Potassium perchlorate, percent	14.0	14.0	14.0
Zirconium-nickel alloy, powdered Type I 70/30 percent	26.0	9.0 ± 2.0[a]	3.0 ± 1.0[a]
Zirconium-nickel alloy, powdered Type II 30/70 percent	------	17.0 ± 2.0[a]	23.0 ± 1.0[a]

[a] The proportions of the fuel ingredients may be varied within the tolerances specified to meet the burning rate requirement.

3.2.2 For purposes of analysis, delay composition, when tested as specified in 4.3, shall conform to the following composition:

Component	Type I	Type II	Type III
Barium chromate, percent	59.0 ± 2.0	59.0 ± 2.0	59.0 ± 2.0
Potassium perchlorate, percent	14.0 ± 2.0	14.0 ± 2.0	14.0 ± 2.0
Total zirconium, percent	18.2 ± 1.2	11.4 ± 2.0	9.0 ± 1.6
Total nickel, percent	7.0 ± 1.2	13.8 ± 2.0	16.2 ± 1.6

3.3 **Moisture.** The moisture content shall be 0.04 percent maximum and shall be determined as prescribed in paragraph 4.4.

3.4 **Functioning.** The delay composition shall have an average burning rate in seconds per inch and standard deviation in burning rate as follows:

TABLE I

	Type I	Type II	Type III
Average burning rate in seconds per inch	2.0 ± 0.5	5.0 ± 0.5	12.0 ± 2.0
Standard deviation of burning rate maximum, seconds, (based on 25 replicates)	0.10	0.14	0.3

A lot of delay compostition will be considered to have complied with these requirements if samples selected, loaded into delay elements, and tested in accordance with 4.6 comply with the acceptance provisions specified herein.

3.5 **Workmanship.** The mixing of the ingredients in each batch and the blending batches of the delay composition shall be of such quality as to give homogeneity to the lot.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facifities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions in 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractors system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time of the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with the subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification*. Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with (a) or (b) below as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector, and depending on the nature of the deviation, may consist of lot rejection, screening, resampling, reinstruction of the suppliers employees, or other appropriate action:

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance*. Surveillance shall include, but is not limited to-

(a) Observation of procedures concerning lot formation and identification.

(b) Observation of sampling procedures and application of acceptance criteria.

(c) Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or of approved equivalents thereto.

(d) Review of procedures for control and disposition of nonconforming material.

4.1.3.2 *Product inspection*. Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provision of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractors records are reliable.

4.2 **Inspection provisions**

4.2.1 *Lot formation*. A lot shall consist of one or more batches of Delay Composition produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of Delay Composition that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with Standard MIL-STD-105 or MIL-STD-1235, as applicable. The criteria and

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procedures for the assignment of inspection lot interfix numbers shall be in accordance with Standard MIL-STD-1168.

4.2.2 *Testing.*

4.2.2.1 *Sampling.* Five primary samples of approximately 8 ounces each shall be selected by the inspector so as to be representative of the lot. Each primary sample shall be selected from a different container unless there are fewer than five containers in the lot in which case at least one sample shall be taken from each container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 12 ounces shall be made from equal portions of the primary samples. The composite samples shall be thoroughly mixed and placed in an airtight container, labeled to show the name of the material, manufacturer, plant, contract or purchase order number, lot number, and lot size. The functioning test shall be made on each primary sample and all other acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with all the requirements of this military specification. All primary samples shall be held for possible future examinations should the composite sample fail to comply with the requirements.

4.3 **Determination of composition**

4.3.1 *Potassium perchlorate.* Transfer an accurately weighed portion of approximately one gram of the sample of a tared 9.0-cm. No. 42 Whatman filter paper, or substantial equal. Leach with 10-ml. portions of hot water, transfer the filter cone and residue to a watch glass and place in a drying oven at 100° to 105° C for two hours. Cool in a desiccator and weigh the filter cone and residue, using the blank filter cone as a tare. Reserve the filter cone and residue for the nickel determination. Calculate the loss in weight of the sample to percent potassium perchlorate as follows:

$$\text{Percentage of Potassium Perchlorate} = \frac{99.34A}{W}$$

where:

A = loss in weight of the sample

W = weight of sample on a moisture-free basis

The filter paper shall be folded and tared against a second similarly prepared filter paper which serves as a blank; that is, the filter paper shall be washed with water and dried in accordance with the procedure given above.

4.3.2 *Nickel.* Transfer the filter paper and residue from the potassium perchlorate determination to a 250-ml. beaker. Add 40 ml. of 6N (1:1) hydrochloric acid, cover the beaker with a watch glass and boil the contents of the beaker gently for 2 hours on a hotplate (adding additional 6N hydrochloric acid occasionally to maintain a volume of approximately 40ml). Digest on a steam bath for a minimum of 2 hours, make up to a volume of approximately 40 ml. with 6N hydrochloric acid and allow to remain at room temperature overnight. Filter through a No. 42 Whatman filter paper, collecting the

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filtrate in a 250-ml. volumetric flask. Wash the residue with approximately 150 ml. of water and make up to volume. Discard the residue. Pipette a 25-ml. aliquot into a 600-ml. beaker and dilute with water to approximately 250 ml. Add 5 grams of tartaric acid and 10 ml. of a 10-percent solution of ammonium chloride. Add concentrated ammonium hydroxide until the solution is slightly alkaline to litmus. If a precipitate forms, add several drops of 12N hydrochloric acid to clear the solution and again neutralize with concentrated ammonium hydroxide. Transfer the beaker to a hotplate, heat to nearly boiling and add, by means of a pipette, 10 ml. of a 1-percent solution of dimethyl-glyoxime-inethanol with vigorous stirring. Add concentrated ammonium gydroxide until the mixture has a distinct ammoniacal odor. Place on a steam bath for approximately 1/2 hour and filter with the aid of a tared, medium porosity, sintered-glass crucible. Wash with three 20-ml. protions of 47.5 percent (1:) ethanol, dry in an oven at 110° to 120° C for 2 hours, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent nickel as follows: Code No. 02001.

$$\text{Percentage of Nickel} = \frac{20.32A}{W}$$

where:

A = increase in weight of the crucible

B = weight of sample represented by the aliquot taken (calculated on a moisture-free basis)

4.3.3 *Barium chromate*. Pipette a 100-ml. aliquot of the solution originally obtained in the nickel determination to a 400 ml. beaker and dilute with water to approximately 250 ml. Heat to boiling on a hotplate and add slowly from a 50-ml. pipette 50 ml. of 3N (1:8) sulfuric acid. Allow the mixture to boil gently for 5 minutes and finally to digest for at least one hour on a steam bath. In case of dispute or noncompliance, allow to stand overnight at room temperature. Collect the precipitated barium sulfate in a tared Gooch crucible which has been previously heated in a muffle furnace at 800° ± 50° C for 30 minutes. Wash twice with 0.12N (1:300) sulfuric acid followed by ten 25-ml. portions of hot water. Dry in an oven at 100° to 105° C for 1 hour and then in a muffle furnace at 800° ± 50° C for 30 minutes. Cool in a desiccator and weigh. Calculate to percentage of barium chromate in the sample as follows: Code No. 03001.

$$\text{Percentage of Barium Chromate} = \frac{108.5A}{W}$$

where:

A = increase in weight of Gooch crucible

W = weight of sample represented by the aliquot taken (calculated on a moisture-free basis)

4.3.4. *Zirconium*. Transfer an accurately weighed portion of approximately 0.5

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gram of the sample to a clean 250-ml. beaker. Add 50 ml. of concentrated hydrochloric acid and 25 ml. of concentrated nitric acid. Cover the beaker with a watch glass and add dropwise, with caution, 25 ml. of concentrated sulfuric acid. Transfer the beaker and contents to a hotplate, boil gently until white sulfur trioxide fumes appear, and continue the heating procedure for a minimum of 2 hours. Cool to room temperature and add cautiously under the watch glass sufficient distilled water to make a volume of approximately 150 ml. Stir the mixture thoroughly with a glass rod and allow to cool at room temperature. Filter the precipitated barium sulfate using a Whatman No. 42 paper (or substantial equal), collecting the filtrate in a clean 400-ml. beaker. Wash the residue on the filter paper with sufficient distilled water to make the total volume in the 400-ml. beaker approximately 200 ml. Discard the filter paper and cool the beaker in an ice bath to 10° ± 2° C. To the beaker in the ice bath, add, with stirring, 32 ml. of freshly prepared cupferron solution (dissolve 6 grams of cupferron in 100 ml. of cold water and filter.) Filter the precipitate using a Whatman No. 40 paper (or substantial equal) and wash with five 20-ml. portions of cold 1N (1:10) hydrochloric acid. Transfer the precipitate and paper to a tared crucible and dry in an oven at 100° to 105° C for 2 hours. Place the crucible and contents in a muffle furnace at 300° to 400° C until complete charring of the paper has occurred and the rush of gases from decomposition of the organic matter has ceased. Increase the temperature to 900° ± 50° C and continue the ignition for 1/2 hour. Cool in a desiccator and weigh as zirconium oxide. Code No. 04001.

$$\text{Percentage of Zirconium} = \frac{74.03A}{W}$$

where:

A = gain in weight crucible

W = weight of sample on a moisture-free basis

4.4 **Determination of moisture.** Transfer an accurately weighed portion of approximately 2 grams of the sample to a tared shallow receptacle and dry in an oven for 2 hours at 100° ± 5° C. Cool in a desiccator and weigh. Calculate the loss in weight of the receptacle and contents to percent moisture in the sample. Code No. 05001.

4.5 **Determination of average particle diameter (applicable only to potassium perchlorate).** The average particle size shall be determined in accordance with Standard MIL-STD-1233, Method 100. Code No. 06001.

4.6 **Functioning test** (see 6.2)

4.6.1 *Procedure*. From each primary sample, load a group of 25 delay assemblies as indicated on Dwg. 7548570. The height of column shall be adjusted to 1.00 ± 0.04 inch. Assembly delay element into Fuze, Grenade, Hand, M204A2, without detonator, and condition for 16 hours at a temperature of 70° ± 5° F. Fire, and record the burning time, using automatic timing equipment accurate to 0.01 second. The fixture or adapter, which is directly in contact with the fuze when it is mounted for firing, should be at a temperature of from 60° to 100° F and all other conditions should be such that the temperature of the fuze, at the time of firing will be 70° ± 5° F. Calculate the average burning time and standard deviation of the burning time for each group of 25 fuzes containing delay composition from the same primary sample and for all fuzes. Calculate

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the average burning time in seconds per inch from the measured column heights and the average burning times for each group of 25 fuzes containing delay composition from the same primary sample and for all fuzes. The methods of calculation to be used are as follows: Code No. 07001.

(1) Average burning time

$$A = \frac{S}{n}$$

where:

A = average burning time in seconds

S = sum of the individual functioning times in seconds

n = number of individual functioning times

(2) Average column height

$$B = 1.263 - \frac{T}{P}$$

where:

B = average column height in inches

T = sum of the individual depths to the top of the delay column in inches

P = number of individual depths

(3) Average burning time in seconds per inch

$$C = \frac{A}{B}$$

where:

C = average burning time in seconds per inch

A = average burning time in seconds

B = average column height in inches

(4) Standard deviation

$$S = \frac{U - D}{n - 1}$$

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where:

S = standard deviation in seconds

U = sum of the squares of the individual functioning times in seconds

$$D = \frac{(S)2}{n}$$

S = sum of the individual functioning times in seconds

n = number of individual functioning times

4.6.2 *Acceptance*. The lot of delay composition shall be considered acceptable provided the average burning time in seconds per inch of each group of 25 fuzes containing delay composition from the same primary sample and of all fuzes is as specified in table I of 3.4 and provided the standard deviation of the burning time of each group of 25 fuzes containing delay composition from the same primary sample and of all fuzes does not exceed the minimum specified in table I of 3.4: If one group of 25 fuzes, containing delay composition from the same primary sample, fails to comply with the requirements of table I of 3.4, a retest of this primary sample shall be permitted using 50 fuzes. If the combined results from the original test and retest comply with the requirements in table I of 3.4, the lot of delay composition shall be considered acceptable; if they fail to comply with the requirements of table I of 3.4, the entire lot of delay composition shall be rejected subject to correction and resubmission for retest. If two or more groups of 25 fuzes, each group containing delay composition from a single primary sample, or if all fuzes fail to comply with the requirements in table I of 3.4, the entire lot of delay composition shall be rejected subject to correction and resubmission for test.

4.7 **Resubmission and retest.** If the composite sample, or any primary sample subjected to test, fails to pass the chemical test, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from any or all of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. A rejected lot of delay composition may be reworked once, but no additional quantities of the ingredients shall be added during reworking except those necessary to increase or decrease the sum of the percentages of the oxidizing agents (barium chromate and potassium perchlorate) and increase or decrease the sum of the percentages of the fuel ingredients (the alloys) so that the ratio of the two sums is 74:26. The resubmitted lot shall be accepted, provided new samples, selected in accordance with 4.2, pass all the tests required by this specification.

5. PREPARATION FOR DELIVERY

5.1 **Packing.** Delay composition shall be packed in metal containers, of no more than 150 pounds, unless otherwise specified by the procuring agency, in such a manner as to comply with the applicable regulations of the Interstate Commerce Commission (see Code of Federal Regulations 49 CFR 71 - 90) and as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

6. NOTES

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6.1 **Ordering data.** Procurement documents shall specify the title, number and date of this specification.

6.2 If the item of ammunition, in which the entire lot of delay composition is going to be used, is known, it is permissable to use this item of ammunition to test the functioning and acceptance of the delay composition instead of the M204A2 hand grenade fuze as specified in 4.7. The test shall be conducted in the known item. Loaded as shown on the applicable drawing, and must meet the requirement of the applicable specification.

6.3 **Inspection code numbers.** The five digit code numbers assigned to the inspection items herein are to facilitate future data collection and analysis by the Government.

Custodian:

Army - MU

Preparing activity:

Army - MU

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